

# Problem Set I

Econometrics I - FGV EPGE  
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## Problem 1 – (points: 1)

Let  $X$  be a scalar random variable with density  $f(x)$ . Let  $K(\cdot)$  be a symmetric second-order kernel. For a given point  $x$  in the interior of the support of  $f(\cdot)$ , define the density estimator as

$$\hat{f}_n(x) \equiv \frac{1}{nh} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right),$$

where  $h > 0$  is a bandwidth parameter and  $X_1, \dots, X_n$  are independent and identically distributed (i.i.d.) random variables with density  $f(\cdot)$ .

This exercise will show you how to ensure that  $\hat{f}_n(x) \xrightarrow{p} f(x)$  as  $n \rightarrow \infty$ ,  $h \rightarrow 0$ , and  $nh \rightarrow \infty$ . This is the same asymptotic framework as in the slides.

- a) Show that  $\hat{f}_n(x) \geq 0$  for all  $x$ , and that  $\int_{-\infty}^{\infty} \hat{f}_n(x) dx = 1$  for all  $n$ .
- b) Assume from now on that  $f$  is continuous at  $x$ . Show that  $\mathbb{E}[\hat{f}_n(x)] = f(x) + o(1)$ .
- c) Show that  $\text{Var}(\hat{f}_n(x)) = \frac{1}{nh} \cdot f(x)R(K) + o\left(\frac{1}{nh}\right)$ , where  $R(K) = \int_{-\infty}^{\infty} K^2(u) du$ .
- d) Argue that these results imply that  $\hat{f}_n(x)$  is consistent for  $f(x)$ .
- e) Now, assume that  $f$  is twice continuously differentiable at  $x$ . Show that

$$\mathbb{E}[\hat{f}_n(x)] = f(x) + \frac{h^2}{2} f''(x) + o(h^2)$$

- f) Explain in words how the local convexity of  $f$  might (or might not) affect this finite-sample bias.

Hint: A very useful resource for this question is Chapter 17 from *Probability and Statistics for Economists* by Bruce Hansen.

## Problem 2 – (points: 1)

Consider the nonparametric regression model:

$$Y_i = m(X_i) + u_i, \quad i = 1, \dots, n$$

where  $\mathbb{E}[u_i|X_i] = 0$  and  $m(\cdot)$  is an unknown smooth function. The **local-linear estimator** of  $(m(x_0); m'(x_0))$  at a point  $x_0$  is defined as the solution to the weighted least squares problem:

$$\{m_{LL}(x_0), m'_{LL}(x_0)\} \equiv \arg \min_{\alpha, \beta} \sum_{i=1}^n K_h(X_i - x_0) [Y_i - \alpha - \beta(X_i - x_0)]^2$$

where  $K_h(u) = \frac{1}{h} K\left(\frac{u}{h}\right)$  is a second-order continuous kernel function with bandwidth  $h > 0$ . You have a random sample of size  $n$ :  $\{(Y_1, X_1), \dots, (Y_n, X_n)\}$ . We also assume that  $X_i \in \mathbb{R}$ .

- a) Show that the local-linear estimator can be written as a Generalized Least Squares (GLS) estimator. Specifically, show that there exists an  $n \times 1$  vector  $V$ , an  $n \times 2$  matrix  $Z$ , and an  $n \times n$  matrix  $W$  such that:

$$\begin{pmatrix} m_{LL}(x_0) \\ m'_{LL}(x_0) \end{pmatrix} = (Z'W^{-1}Z)^{-1}Z'W^{-1}V$$

- b) What happens to this estimator if, for a given  $n$ , we have that  $h \rightarrow \infty$ ?
- c) When deriving the asymptotic theory for the local-constant estimator, we assumed the bandwidth was such that  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$ . Explain intuitively why each of these two conditions is necessary.
- d) Are such conditions also needed for the asymptotic theory of the local-linear estimator? What are the main differences between these two estimators? Briefly explain.

### Problem 3 – (points: 3)

This is an empirical question based on Karlan and Zinman (2008), “[Credit Elasticities in Less-Developed Economies: Implications for Microfinance](#)”, *American Economic Review*. The data, `karlan.csv`, is in the `problem_sets/PS1` folder of the class GitHub repo.

- a) What is the main research question in the paper? What is the most striking finding? Answer in just a few sentences.
- b) Your goal will be to estimate  $\mathbb{P}(\text{applied} = 1 | \text{offer4} = x)$ . Note that `applied` is a binary variable, while `offer4` is continuous. These are the only variables you will need. Notice that

$$\mathbb{P}(\text{applied} = 1 | \text{offer4} = x) = \mathbb{E}[\text{applied} | \text{offer4} = x]$$

- c) Use a Gaussian kernel to estimate this probability and show a plot of your estimates for a range of values of `offer4`. Do this with three different bandwidths:

- Silverman's rule of thumb:  $h = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5}$ , where  $\hat{\sigma}$  is the standard deviation of offer4 and  $n$  is the number of observations;
- A value much *smaller* than that;
- A value much *larger* than that;

- d) Do the same with the Epanechnikov kernel, using the same bandwidths as before.
- e) Compare the results of the two kernels qualitatively.

## Problem 4 – (points: 1)

Let  $X_i, i = 1, \dots, n$  be an i.i.d. sample of observations with distribution  $U[0, \theta]$ . A natural estimator (in fact, the MLE) of  $\theta$  is  $X_{(n)}$ , where

$$\min\{X_1, \dots, X_n\} = X_{(1)} \leq \dots \leq X_{(n)} = \max\{X_1, \dots, X_n\}$$

denote the ordered values of the data. These statistics are sometimes referred to as the order statistics of the data. Consider the following root:

$$R_n = n(X_{(n)} - \theta),$$

- a) Show that  $R_n \leq 0$ .
- b) Let  $J_n(x, P) = P(R_n \leq x)$  be the cdf of the root  $R_n$  above. Show that  $J_n(x, P)$  converges in distribution to  $J(x, P) = P(-\theta X \leq x)$ , where  $X \sim \exp(1)$ , i.e.,

$$J(x, P) = \begin{cases} e^{x/\theta}, & x \leq 0 \\ 1 & \text{otherwise} \end{cases}$$

Hint: for every  $r \in \mathbb{R}$ , remember that  $(1 + \frac{r}{n})^n \xrightarrow{n \rightarrow \infty} e^r$ .

- c) We will now prove that the bootstrap will not work in this case. Consider a non-random sequence of probability distributions  $\{P_n\}_{n=1}^\infty$  such that each distribution  $P_n$  puts equal mass on  $n$  distinct points in the interval  $[0, \theta]$ .

For each  $n$ , let  $\{X_{i,n}\}_{i=1}^n$  be an i.i.d. sample from  $P_n$ . Denote by  $X_{(n),n}$  the maximum of this  $n$ -th sample. Also, denote by  $\theta(P_n)$  the maximum of the support of  $P_n$ .

Now, show that  $\lim_{n \rightarrow \infty} P(n(X_{(n),n} - \theta(P_n)) \leq -\epsilon) \leq e^{-1}, \forall \epsilon > 0$ .

- d) Use this result to show that  $J_n(x, P_n)$ , which is the distribution of the bootstrapped root under the measure  $P_n$  does *not* converge to  $J(x, P)$ .

Hint: pick  $\epsilon < \theta$ .

- e) Argue, using the last item, that the bootstrap does not work in this case. Think about the relationship between the empirical distribution and the sequence of distributions  $\{P_n\}_{n=1}^\infty$ .

## Problem 5 – (points: 2)

This question is about the bootstrap. Suppose that  $y_i = \alpha + \beta x_i + \epsilon_i$ , where  $\alpha = 2$ ,  $\beta = 5$ ,  $\epsilon_i | x_i \sim N(0, 2 + x_i^2)$ , and  $x \sim N(0, 3)$ .

- Compute the conditional distribution of  $y_i$  given  $x_i$ ;
- Simulate a random sample of size  $n = 100$  and create a scatter plot of your data;
- Estimate the parameters using OLS;
- Report the estimates and three different symmetric confidence intervals for  $\beta$  using different measures of standard errors:
  - The usual OLS standard error that ignores heteroskedasticity;
  - A heteroskedasticity-robust standard error;
  - A bootstrap standard error, using  $B = 1000$  bootstrap samples. You should also plot the histogram of  $\hat{\beta}$  computed from the bootstrap samples.

It will be helpful for the next item if you structure your code so that it receives a certain sample  $\{(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)\}$  and spits out the three different confidence intervals.

- e) Repeat the steps from items b), c), and d) for  $m = 500$  times and check how frequently the three different confidence intervals cover the true value of  $\beta = 5$ . You can use the same code from item b) to do this, but you will need to modify it so that it runs  $m$  times and stores the results.

## Problem 6 – (points: 2)

This question connects the bootstrap with the kernel density estimator from Problem 1. It is mostly about coding and explaining what you find. Let  $X_1, \dots, X_n$  be i.i.d.  $N(0, 1)$ , so the true density is  $f = \phi$ , the standard normal density. At a point  $x_0$ , estimate  $f(x_0)$  with

$$\hat{f}_n(x_0) \equiv \frac{1}{nh} \sum_{i=1}^n \phi\left(\frac{X_i - x_0}{h}\right),$$

a kernel density estimator with a Gaussian kernel and Silverman's rule of thumb,  $h = 1.06 \cdot \hat{\sigma} \cdot n^{-1/5}$ , where  $\hat{\sigma}$  is the sample standard deviation.

- a) Write a function that receives a sample  $\{X_1, \dots, X_n\}$  and a point  $x_0$ , and returns a 95% bootstrap confidence interval for  $f(x_0)$  based on the root  $R_n = \hat{f}_n(x_0) - f(x_0)$ . Specifically:
- Draw  $B = 499$  bootstrap samples (with replacement) from the data;
  - In each bootstrap sample, recompute  $\hat{f}_n^*(x_0)$ , using the *same* bandwidth  $h$  as in the original sample;
  - Let  $q_{0.025}$  and  $q_{0.975}$  be the 2.5% and 97.5% quantiles of the  $B$  values of  $\hat{f}_n^*(x_0) - \hat{f}_n(x_0)$ , and report the interval  $[\hat{f}_n(x_0) - q_{0.975}, \hat{f}_n(x_0) - q_{0.025}]$ .
- b) Set  $n = 500$ . Simulate  $M = 500$  samples and, for each one, compute the confidence interval from item a) at  $x_0 = 0$  and at  $x_0 = 1$ . Report how often each interval covers the true value  $f(x_0)$ .
- c) Explain the difference in coverage between  $x_0 = 0$  and  $x_0 = 1$ .
- Hint: compute  $f''(x)$  for the standard normal density and recall item e) from Problem 1.
- d) Let  $\mathbb{E}^*[\cdot]$  denote the expectation over bootstrap samples, holding the original data fixed. Show that  $\mathbb{E}^*[\hat{f}_n^*(x_0)] = \hat{f}_n(x_0)$ . What does this tell you about the ability of the bootstrap to detect the bias of  $\hat{f}_n(x_0)$ ?
- e) Redo item b) at  $x_0 = 0$  with  $n = 2,000$ . Does more data fix the coverage? Explain why, comparing how the bias and the standard deviation of  $\hat{f}_n(x_0)$  shrink with  $n$  when  $h$  follows Silverman's rule.
- f) Go back to  $n = 500$  and  $x_0 = 0$ , but now use a smaller bandwidth,  $h/3$ , both in the original sample and in the bootstrap samples. What happens to the coverage and to the average width of the confidence interval? Explain the trade-off.